

**Group Design Practical
Briefing Document Hilary Term 2026**

All second-year undergraduates reading Computer Science, Mathematics and Computer Science, or Computer Science and Philosophy take part in a group design exercise in their second year.

Goal

The goal of the group design practical is to give you, the student, a chance to understand and learn the skills required to succeed in delivering a real-world project, not just writing a program.

In addition to using the skills learnt during the four core programming courses to actually code, to succeed in this project you'll need to

- engage with your customer to understand what their needs are
- agree with the customer what your solution will and won't do in order to solve their needs (and update/negotiate changes with them as necessary)
- self-organise with your team to distribute work *and* make sure the whole solution works together
- plan your time to deliver what the customer needs when they need it
- provide the solution to the customer in a form
 - o they can easily deploy (installation, configuration)
 - o they can easily use (documentation, briefings, maintainability, good diagnostics)
 - o they won't break (careful testing).
 - o and considers the ethical and societal impacts of the solution.

These are all solid basics that any tech company or research group expects of any good contributor – this is your chance to develop these skills as part of your degree.

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Key Dates

Briefing Meeting	Wednesday 21 st January, 1pm	Week 1
OOP Lecture 1	Wednesday 21 st January, 2pm	Week 1
OOP Lecture 2	Wednesday 28 th January, 1pm	Week 2
OOP Lecture 3	Wednesday 28 th January, 2pm	Week 2
Deadline to fix team members	Monday 2 nd February, 12pm	Week 3
OOP Lecture - AI	Wednesday 4 th February, 1pm	Week 3
OOP Lecture 4	Wednesday 4 th February, 2pm	Week 3
Confirmation of project choices	Friday 6 th February, 12pm	Week 3
Briefing Meeting on specific projects		Week 4
Responsible Innovation Seminar	Wednesday 11 th February, 2pm	Week 4
Deadline to fix projects	Thursday 12 th February, 12pm	Week 4
Project Co-ordination and Management Seminar	Wednesday 18 th February, 2pm	Week 5
First progress meeting with external (and internal) supervisor – agree customer requirements and project specification and plan	Week 6 - Time and day to be confirmed by team and supervisor	Week 6
Individual mid-Term meeting with internal supervisor		Week 7
Software version control seminar	Wednesday 4 th March, 2pm	Week 7
Software testing seminar	Wednesday 11 th March, 2pm	Week 8
Second progress meeting with external (and internal) supervisor – Progress report, initial implementation and test results	Week 8 - Time and day to be confirmed by team and supervisor	Week 8
Third progress meeting with external (and internal) supervisor – delivery and final report	Trinity Term Week 2 - Time and day to be confirmed by team and supervisor	Week 2
Presentation event	Wednesday 13 th May	Week 3

Organisation

You will be allocated to a team of around 6 people before the Briefing Meeting. Swaps between teams are allowed until the deadline to fix team members – to confirm a swap both students involved should email student-projects@cs.ox.ac.uk.

Teams and projects will be allocated by a random selection process. Each team will be randomly paired with a project from the available list and assigned a supervisor accordingly. Once allocations are complete, teams will be notified of their assigned project and supervisor. Teams may subsequently swap projects with another team, provided that both teams and all individual team members involved agree to the swap. Further details about the project, can be found from page 6 of this document.

You will have an internal and an external supervisor.

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The external supervisor will act as a customer with whom the team engage to understand any detailed requirements for the project and to whom the team deliver their finished product.

The internal supervisor will act as a mentor, ready to help the team deliver their project.

They will arrange an individual mid-term meeting with each member of their team. This is so the student is able to discuss any concerns, the progress and contributions of individual students and the interaction with the industrial supervisor. The meeting should be arranged in week 7, once the first progress meeting has happened and before the second progress meeting. This meeting should be the default option and students may wish to opt out if they wish to do so.

It is the team's responsibility to drive the project, requesting help from the external and internal supervisor as necessary.

As part of the project, each team *must* do the following.

- Arrange an initial planning meeting, with their external (and internal) supervisor in Week 6 of Hilary Term. At this meeting the team should agree what they will be delivering, when, to the customer and agree with their mentor a plan for how they'll achieve that.
- Arrange a progress report meeting with their external (and internal) supervisor in Week 8 of Hilary Term. At this meeting, they'll present a progress report to the customer on their module implementations, test plans and test results. If the team and their internal and external supervisor agree, the team may instead provide a brief written report on their progress by the end of Hilary Term.

This meeting is also a good point to negotiate changes to the scope or timeline for what they are delivering (if for example it turns out that their project is significantly harder than originally expected, they can negotiate to reduce the scope). It is absolutely fine for teams to agree with the customer to change the scope of what they're delivering as long as the customer has reasonable warning (while it is absolutely not fine for example for the team to propose a reduction in scope 24 hours before the final delivery date!)

- Arrange a delivery meeting with their external (and internal) supervisor in Week 2 of Trinity Term. At this meeting, they deliver the agreed project to the customer, including the product, installation instructions, documentation, etc. as previously agreed.
- Arrange separate progress meetings with their internal supervisor unless the internal supervisor is able to attend progress meeting with the external supervisor. Each group should meet their internal supervisor at least three times.

In Week 2 of Trinity Term, each group will deliver to the internal supervisor a final project report and each student will also deliver to the internal supervisor a one-page summary of their individual contribution.

Teams are encouraged to engage with their external and internal supervisors more frequently than these three mandatory meetings.

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On Wednesday 13th May there will be an event to celebrate you and your hard work. Each group will be able to present their work and give a 5-minute talk, and there will be cash-prizes, too!

You'll receive detailed information nearer the time.

Overview and Marking

The group practical exercise is designed to take 20-30 hours of student time, mainly during Hilary Term (so a team of 6 delivers a project of 120-180 hours of work). The contribution of each student will be based on the group report, demonstration, and individual contribution sheet. This mark will count for approximately one third of the practical marks for the second year. Students must pass the group practical in order to pass the practical component of the course. The group design exercise will be marked on a scale of S-, S(pass), S, S+. These marks should be converted to a numerical mark using the following scale:

S+	100
S	60
S (pass)	40
S-	20

This exercise gives students experience of working against rigid deadlines, with a team of colleagues not of their own choosing, using externally supplied tools to undertake a fixed project. This will give some idea of the problems encountered in normal professional practice. Groups will be expected to exhibit professional skills in design, quality and management. Specifically, they will have to show that the work has been carefully planned, that components and systems have been properly tested, and that members of the group have cooperated effectively. The review meetings with mentors provide an opportunity to monitor group progress and for general discussion, but mentors will not be expected to provide technical advice or resolve technical issues. The responsibility for organising and completing the work lies squarely with the group members.

Our intention is that all group projects should be successful, and all students are expected to contribute to their own project in accordance with the timetable. Groups often encounter problems, and students are expected to manage these as they arise. However, if serious problems are encountered, such that members of the group are unable to resolve them, students should contact the academic admin team (academic.administrator@cs.ox.ac.uk).

Deliverables

Each part of the project gives rise to a set of deliverables *which must be given to the internal supervisor at the review meetings*. The internal supervisors will forward all group and personal reports to the examiners for inclusion in the practical work portfolio of all group members.

Specification and plan

Project topics are presented in the form of an outline design brief. Part of the work is to undertake a proper requirements analysis for the chosen project. However, it is important not to develop an over-elaborate specification which commits the group to more work than is necessary. The first major task is

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therefore to turn a relatively open and informal design brief into a more detailed project specification, setting out the major components of the system you will produce, what documentation you'll deliver, what diagnostics you'll provide, etc. Once this is agreed with your internal supervisor, you must produce a project plan which sets out who will do what, the time needed for developing and testing each module, dependencies between modules, etc.

When planning and executing the project, it is important to work to a budget. The entire project should take no more than 40 hours per team member, and records should be kept of time invested. Groups should set realistic targets and achieve them; there will be no additional credit for over-elaborate projects, or for individuals who offer or do more than is required. Each member of the group is expected to gain experience of programming in the course of the project – this may possibly involve test harnesses or scripts, data conversion utilities, a tutorial system, external interfaces, demonstration examples, installation packages, or other code and associated materials as appropriate to the project and the individuals in the team.

Module implementation and testing

The next task is to implement the components and test them. This is likely to require the construction of special test harnesses for separate classes. The deliverable for this task is a written *progress report* describing the testing procedures and results.

Project delivery and report

The final task is to piece the whole system together, test it and ensure that it is adequately documented. At the final meeting, the group must:

- Demonstrate the solution to the external supervisor.
- Deliver the solution and associated documentation in a format that the external supervisor can use.
- Submit a brief (approximately 4 pages long) group report to the internal supervisor describing the project aims, outcomes, successes and failures, and any lessons learnt. This report should include a short (approximately half a page) discussion of responsible innovation, exploring (i) the ethical issues you needed to consider during your design and development activities and (ii) the potential societal impacts (both positive and negative, intended and unintended) of the solution that you have developed.

Each team member must also submit a one-page summary of their own individual contribution to the internal supervisor.

Project A: Brain Barge

Squid Software

Inland shipping plays a vital role in container logistics between the seaports and inland terminals. It is the job of human planners to determine which barges visit which terminals, when, and which containers to load or discharge, while maximising transport capacity utilisation and on time performance. Currently, planners have access to all required data but lack intuitive visual cues and hard numbers to aid them in making the right decisions. They must manually combine, sort, and filter information about rotations, containers, vessel capacity, and terminal schedules in various software applications and Excel sheets to get an understanding of the puzzle they need to solve. The challenge is not only to visualise what's happening but to assist in making these decisions. At the end of the day, a planner needs to come up with an efficient plan while keeping all constraints given a set of barges, terminal visits, and containers that need to move within varying time windows. However, even a good plan suffers from reality. When conditions change, such as the closure of a major sea terminal, or a delay due to a broken lock, we need to mitigate the impact by making a new plan. By the end of the project, students will have created a prototype that allows planners to:

- Visually assess barge utilisation and routing choices.
- Understand the trade-offs between different planning scenarios.
- See and adapt to disruptions in near real-time.

The result will not only demonstrate technical and design ability but also show how data visualisation and intelligent algorithms can support complex logistical decision-making in practice.

Aims

- Develop a visualisation that gives clear insight into:
 1. The rotation of each barge (sequence of terminal calls over time).
 2. The loading and discharge per terminal (TEU/ton) and segment.
- Which containers are unassigned, delayed, or risk missing their closing times?
- (Bonus) Design an algorithm that automatically proposes a feasible rotation plan or allocation of containers to barges and terminals. The algorithm should explain its reasoning clearly, so that the planner can validate or adjust it.
- (Extra bonus) Demonstrate robustness to change. For example, simulate a disruption (e.g., a port terminal closes due to a strike causing an idling barge and a spike in capacity the next day) and show:
 1. How this is visualised in the interface.
 2. How your algorithm or interface supports re-planning or mitigation (alternative routes, other terminals, rescheduling).

Deliverables

- A web-based visualisation of barges, rotations, and container movements over time.
 1. A database / API consumes a provided dataset of:
 2. Barges and their properties (capacity, constraints).
 3. Rotations and historical calls from the past week.
 4. Containers to be transported next week (origin, destination, timing, priority).
- (Bonus) An algorithmic component that suggests rotation or assignment decisions, with human-interpretable justification.
- (Extra bonus) A change-handling demo that simulates disruptions and shows their effect on the plan and GUI.

Technology Requirements

- The web GUI preferably written in some Node.JS or TypeScript framework or vanilla JS. Most of Squid's current web interfaces are written in SvelteKit.
- If an API is developed any JS-related language / framework is fine.
- Algorithm may be written in Python, TypeScript, Rust, or Elixir and should run as a standalone service.